DESIGN-TO-SIMULATION DOCUMENTATION

Aerodynamic Analysis of a Precision Guidance Kit (PGK) Fitted to an Artillery Shell

From CAD modelling in SolidWorks to CFD in Ansys Fluent

1. Objective and scope

The aim of this work is to understand how a precision guidance kit (PGK) behaves aerodynamically when it replaces the nose fuze of an artillery shell flying at supersonic speed. The study follows a standard engineering workflow:

- **Design:** model the PGK and the shell in SolidWorks and assemble them.
- **Pre-processing:** build the surrounding air domain and name the boundaries in SpaceClaim.
- **Meshing:** generate a boundary-layer-resolved mesh in Fluent Meshing.
- **Solving:** run a compressible RANS simulation in Fluent.
- **Post-processing and checks:** inspect the flow field, then check convergence of the residuals and of the force and moment coefficients.

Two configurations were simulated. The PGK alone was used first to understand the flow around the nose and fins. The PGK assembled with the artillery shell was then meshed and solved to obtain the force and moment coefficients.

2. Workflow overview

Step	Stage	Tool	Output
1	Part and assembly modelling	SolidWorks	PGK part, shell part, assembly SIH_VEL_v3
2	Geometry clean-up and enclosure	SpaceClaim	Cylindrical air domain with shared topology
3	Meshing	Fluent Meshing	Polyhedral and poly-hexcore mesh with prism layers
4	Solution	Fluent Solver	1000 iterations, converged residuals and force monitors
5	Post-processing	Fluent Results	Pressure, Mach and velocity contours, pathlines
6	Verification and validation	Plots and checklists	Cd, Cl, Cm, roll and Magnus coefficients

3. Design in SolidWorks

3.1 PGK part

The PGK has a rounded nose, a cylindrical body and four flat swept fins placed at 90 degrees around the axis. The fins are modelled as plates with a real thickness, so their leading edges, tips and trailing edges are all captured. A small slot is cut into the body near one fin root, which represents the fin pivot or actuator pocket.

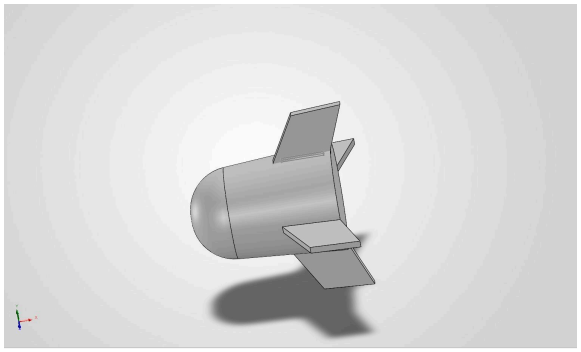


Figure 1. *PGK part model in SolidWorks: dome nose, cylindrical body, four swept plate fins and the fin-root slot.*

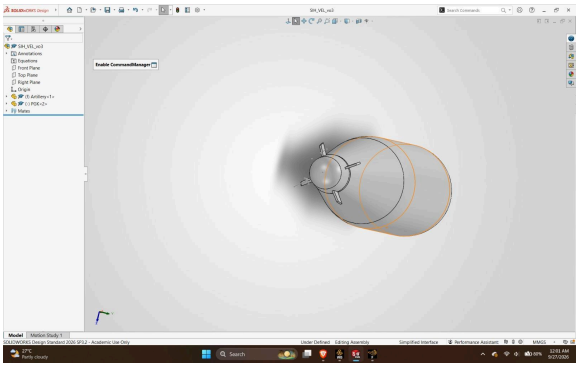


Figure 2. *SolidWorks assembly SIH_VEL_v3 showing the PGK fitted to the artillery shell. The orange outline is the highlighted shell body.*

4. Pre-processing in SpaceClaim

SpaceClaim is used to create the air volume around the model. The project type is Fluid Flow (Fluent with Fluent Meshing). The Enclosure tool wraps the PGK and the shell in a large cylinder, and the Boolean subtraction leaves only the air. The resulting wind tunnel is shown in Figure 6 in Section 5.2.

Parameter	Value in screenshot	Purpose
Enclosure type	Cylinder	Matches the near-axisymmetric flow around a shell
Dimensions shown	3000 mm, 3000 mm and 15000 mm	Places the outer boundaries many body lengths from the shell so they do not disturb the shock
Default cushion	25%	Padding when dimensions are not typed in
Create share topology	Enabled	Keeps faces conformal between bodies
Bodies	Artillery and PGK (2 bodies)	Both are subtracted from the air volume
Named selections	---	Inlet, outlet, far-field, symmetry (if used), wall

5. Flow-field results (PGK alone)

The flow runs from left to right along the x-axis. All contours use the same legends: velocity from 0 to 740 m/s, Mach number from 0.09 to 2.3 and static pressure from -8.49e4 to 4.25e5 Pa.

5.1 Pressure field

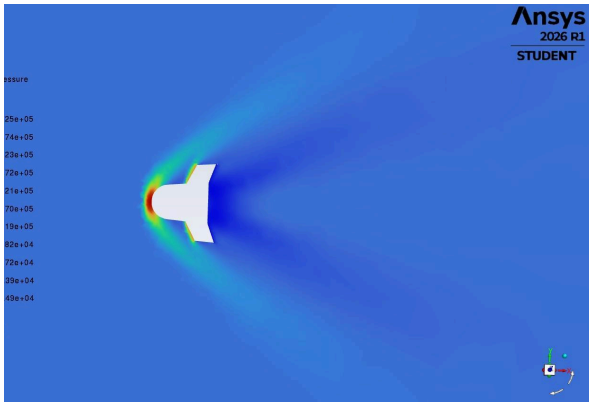


Figure 13. *Static pressure contour on the symmetry plane.*

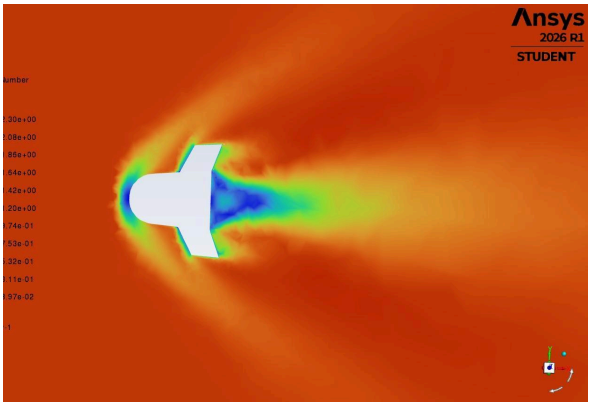


Figure 14. *Mach number contour on the symmetry plane showing the bow shock, fin shocks and subsonic wake.*

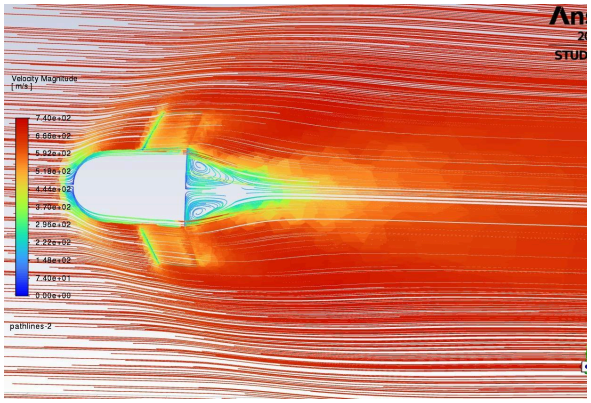


Figure 15. *Pathlines coloured by velocity magnitude on the symmetry plane. Two counter-rotating recirculation bubbles are visible behind the base.*

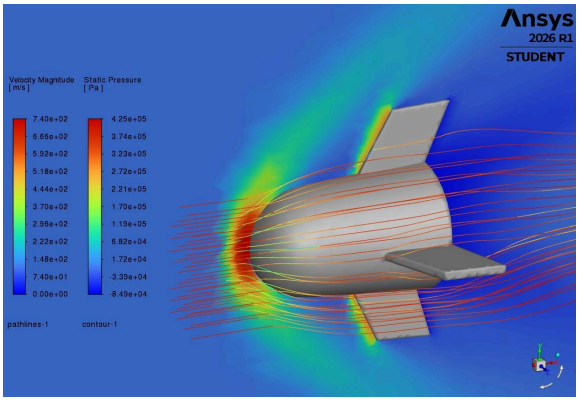


Figure 16. *3D view combining pressure contour and velocity pathlines.*

6. Meshing in Fluent Meshing

The mesh must do three things: resolve the boundary layer on the walls, capture the bow shock and fin shocks in the near field, and resolve the wake behind the base. Everything else can be coarse, which keeps the cell count inside the student licence limit.

6.1 Mesh of the PGK alone

The surface mesh view shows triangles and many thin layers following the fin outline. The size drops sharply near the body and grows toward the far field. We have already seen the results of the wind tunnel above, it is important to first observe and tune the lift and moment coefficient of the PGK, since this would serve as the maneuvering device.

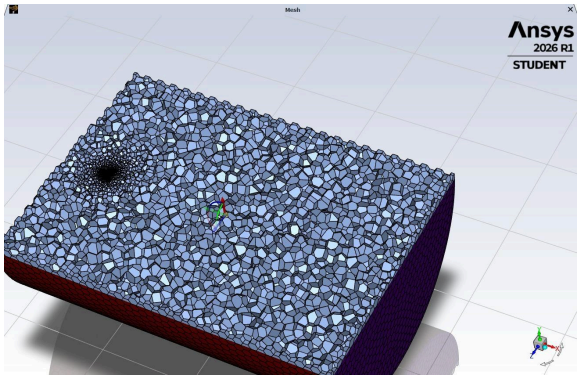


Figure 3. *Polyhedral volume mesh of the PGK-only domain, clipped to show the refinement concentrated around the body (left).*

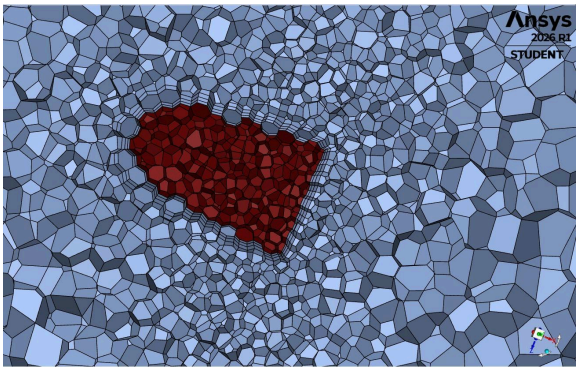


Figure 4. *Close-up of polyhedral cells and prism layers around the PGK cross-section.*

6.2 Mesh of the PGK assembled with the artillery shell

After the PGK-only study, the simulation was repeated on the assembled artillery with the PGK fitted. The wind tunnel used for this case is the cylindrical enclosure from Section 4 wrapped around the full assembly (Figure 6). The mesh of the whole wind tunnel follows.

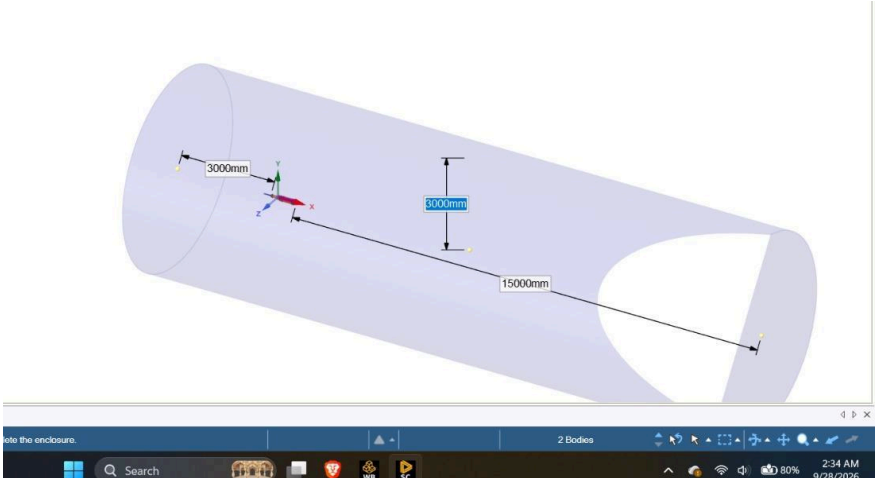


Figure 6. *Wind tunnel (cylindrical enclosure) of the assembled artillery with the PGK fitted, shown in SpaceClaim with the enclosure options.*

For the assembled model the meshing strategy changes to poly-hexcore. Cartesian hexahedral cells fill the bulk of the domain in blocks of increasing size, polyhedral cells form the transition at the outer boundary and near the walls, and prism layers cover the shell and PGK surfaces. The body sits in the left third of the domain, which leaves a long region downstream for the wake.

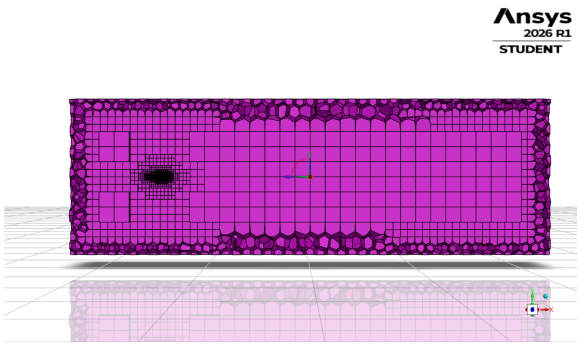


Figure 7. *Cut plane through the whole domain: hex-core blocks of graded size, polyhedral cells at the outer boundary, and the finest cells around the body.*

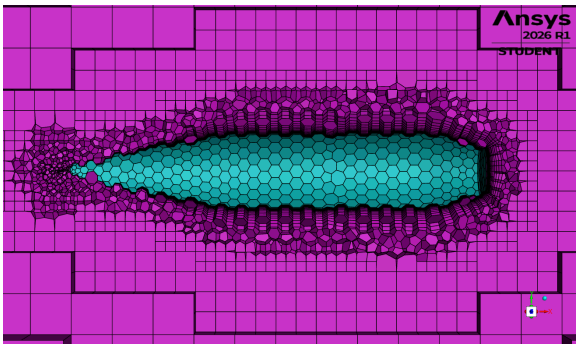


Figure 8. *Zoom on the shell: wall surface mesh (teal), stacked prism layers and the refined wake region behind the base.*

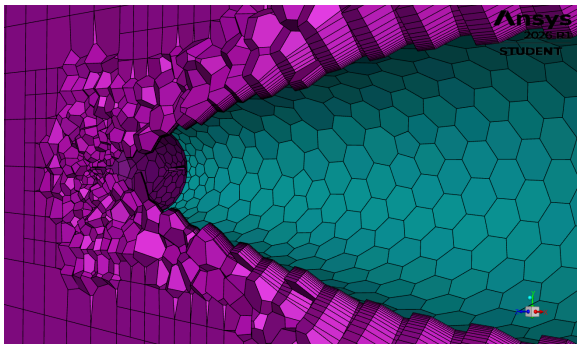


Figure 9. *Nose region: the surface mesh is much finer at the tip, where the bow shock forms and pressure is highest.*

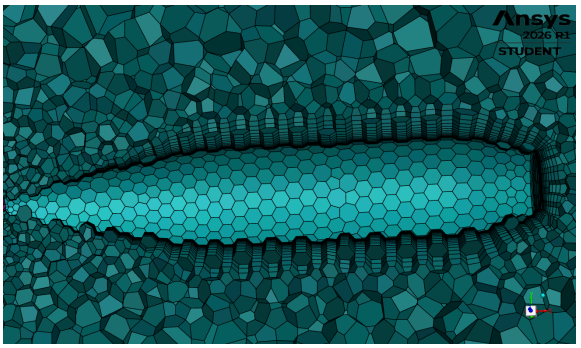


Figure 10. *Prism layers along the shell wall and the abrupt layer end at the flat base.*

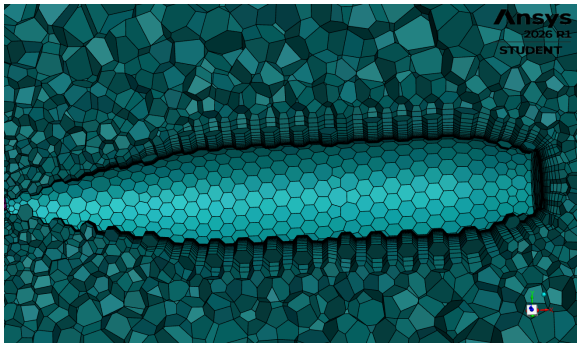


Figure 10. *Prism layers along the shell wall and the abrupt layer end at the flat base.*

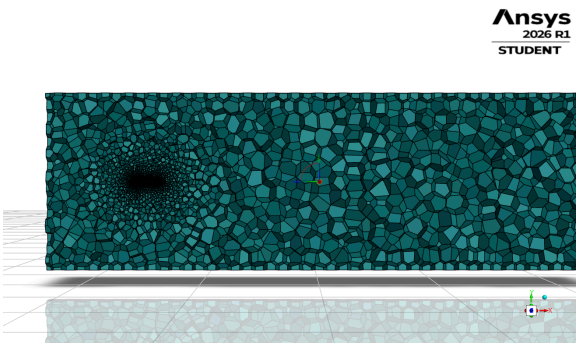


Figure 11. *Outer boundary surface meshed with large polyhedra; cell size drops toward the body.*

6.3 Meshing parameters

Stage	Parameter	What it controls
Surface mesh	Minimum size, maximum size, growth rate, curvature normal angle	Resolves the nose curvature, fin edges and slot
Local sizing	Face size on PGK and fins; body of influence around body and wake	Dense core around the shell and the long wake block
Boundary layer	First layer height, number of layers, growth ratio (about 1.2)	Boundary-layer resolution; target y+ near 1 for k-omega SST
Volume mesh	Poly-hexcore fill, buffer and peel layers, maximum cell length	Cell count and the hex-to-polyhedral transition
Quality	Orthogonal quality, skewness	Solver stability
Actual values	To confirm from the meshing log	Cell count, first layer height and y+ are not visible in the screenshots

7. Solver setup

The monitor and residual plots show that the case solves three velocity components, energy, and turbulence quantities named k and omega. That means a three-dimensional, compressible, k-omega-family turbulence model with the energy equation switched on. The table separates what is visible from what still needs confirming.

Setting	Value	Status
Dimensionality	3D	Seen in x, y and z velocity residuals
Energy equation	On	Seen in energy residual
Turbulence model	k-omega family (k and omega residuals)	Variant (for example SST) to confirm
Flow regime	Peak Mach 2.3, peak velocity 740 m/s	Seen in legends
Pressure range	-8.49e4 to 4.25e5 Pa	Seen in legends; gauge pressure assumed
Iterations	1000	Seen in plots
Solver type and schemes	Density based	Density-based coupled with second-order upwind is typical
Material and viscosity	Air as ideal gas	Air as ideal gas with Sutherland viscosity is typical
Inlet / far-field	Air	Mach number, static pressure, temperature, flow direction
Wall condition	Air	No-slip; shell spin rate if Magnus effects are modelled

Setting	Value	Status
Reference values	---	Reference area and length for the coefficients; moment reference point
Monitors	Cd, Cl, Cm, PGK roll moment, Magnus force, Magnus yaw moment	Seen in Section 9

8. Flow-field results (assembled artillery with PGK)

After the PGK-only study, the flow around the assembled artillery with the PGK fitted was post-processed in the full wind tunnel. The two figures below show the velocity magnitude on a plane through the model and the pathlines around it.

8.1 Velocity magnitude contour

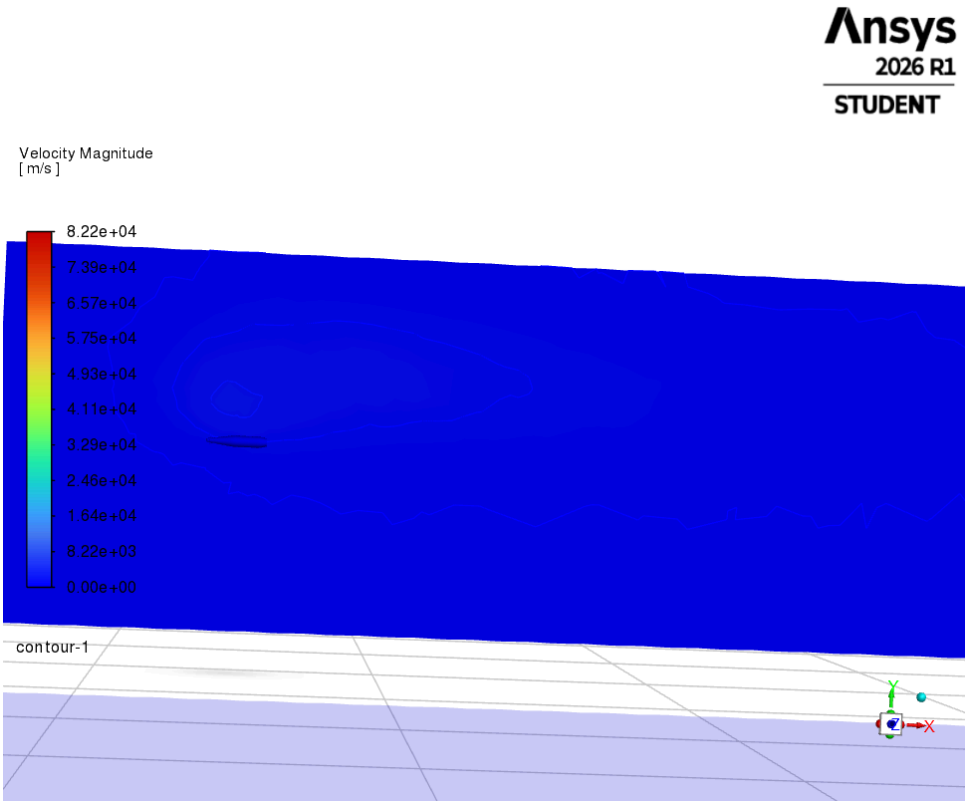


Figure 18. *Velocity magnitude contour on a plane through the assembled artillery and PGK. The colour legend runs from 3 to 8.22e4 m/s, so almost the whole plane appears dark blue.*

The contour is plotted on a plane through the wind tunnel, with the shell visible as a small dark outline on the left. Faint contour lines show a disturbed region around and behind the body. The legend maximum of 8.22e4 m/s is far above the free-stream speed of about 740 m/s seen in the PGK-only study, so a few cells probably carry very large values that stretch the colour scale and hide the real structure. The contour should be re-plotted with the legend range fixed to roughly 0 to 800 m/s before it is used to describe shocks or the wake.

8.2 Velocity pathlines

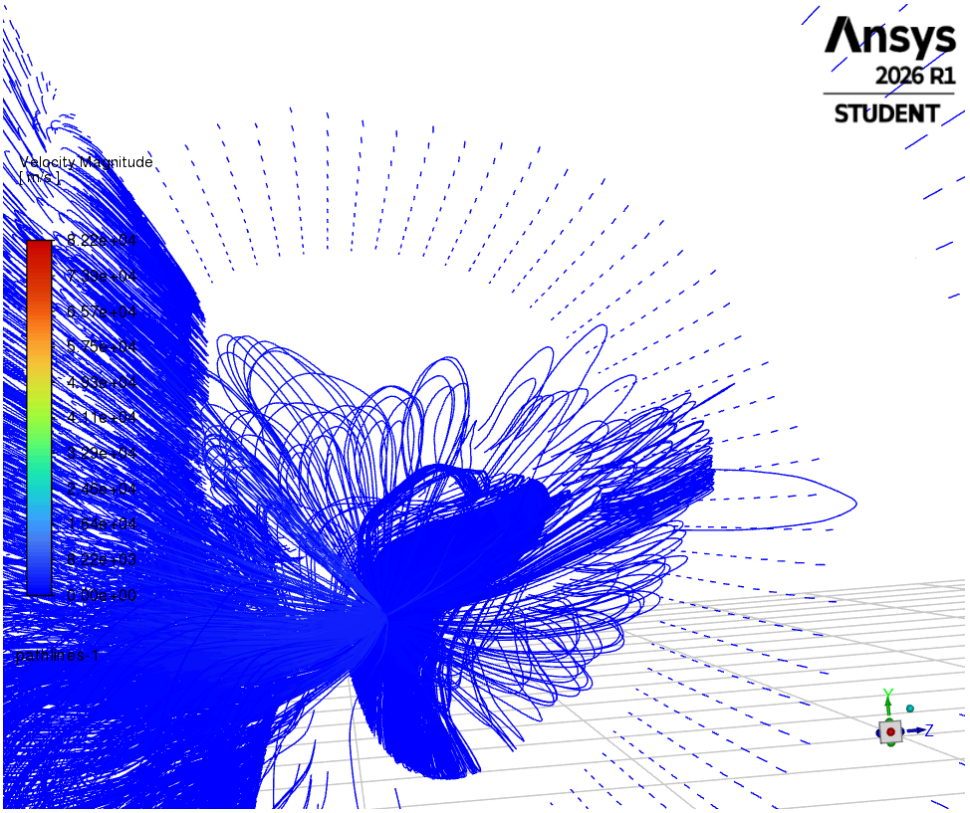


Figure 19. *Pathlines coloured by velocity magnitude around the assembled artillery and PGK, using the same 8.22e4 m/s legend maximum.*

The pathlines form dense bundles and looping curves around the body instead of the smooth, nearly straight lines seen in the PGK-only run. All lines are the same blue because of the 8.22e4 m/s legend maximum, so speed differences cannot be read from this view. Re-plotting with the same fixed legend range and pathlines released from an upstream surface would show the real flow pattern more clearly.

9. Convergence and validation review

9.1 Residuals

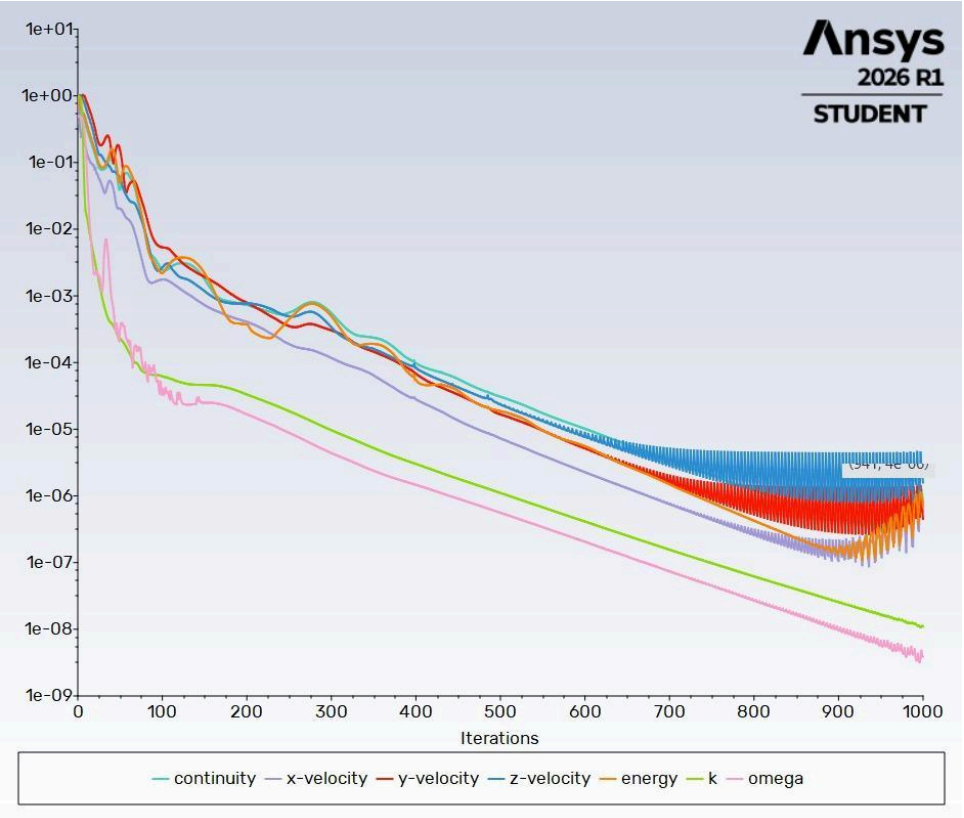


Figure 20. *Scaled residuals over 1000 iterations.*

All residuals fall steadily by several orders of magnitude. By iteration 1000, k is near 1e-8 and omega is near 4e-9. Continuity and x-velocity are around 1e-6 to 1e-7. The y-velocity, z-velocity and energy residuals flatten between 1e-6 and 4e-6 and show a small oscillation after roughly iteration 650. These levels are far below the common 1e-3 criterion. The oscillation is small and probably comes from a weakly unsteady feature such as the base wake or from the solver step size. It is acceptable here because the force coefficients below are flat.

9.2 Force and moment coefficients

Every coefficient has a start-up transient during the first 100 iterations, followed by a flat plateau. This is the usual behaviour when a steady solution is initialised from uniform conditions. All curves are settled by about iteration 300 and stay constant to iteration 1000.

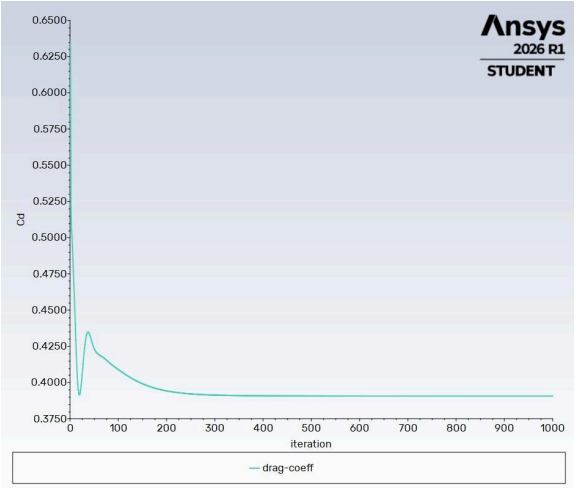


Figure 21. *Drag coefficient history. It drops from about 0.63 at the start and settles near 0.39.*

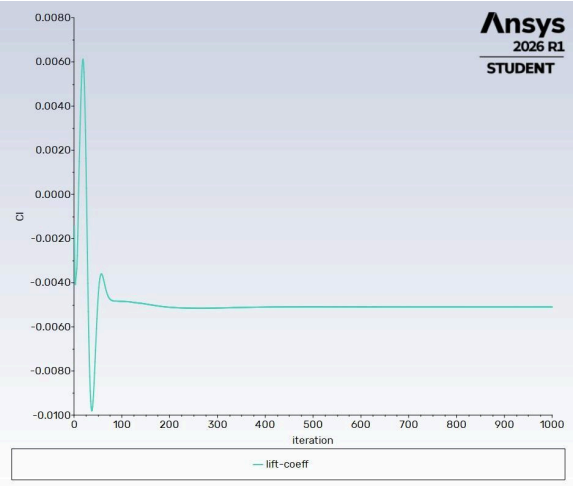


Figure 22. *Lift coefficient history. It settles at a small negative value of about -0.005.*

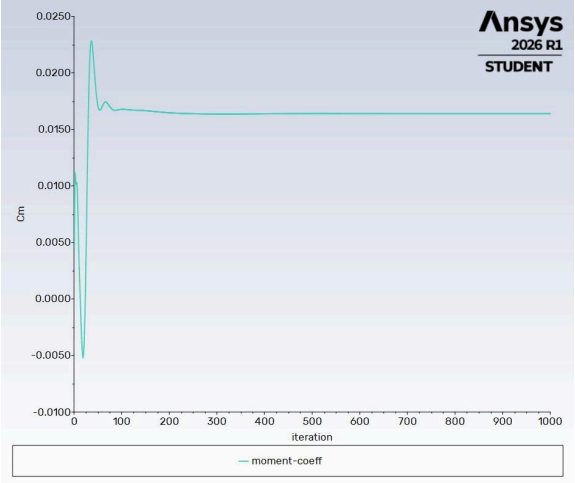


Figure 23. *Pitching moment coefficient history, settling near 0.016.*

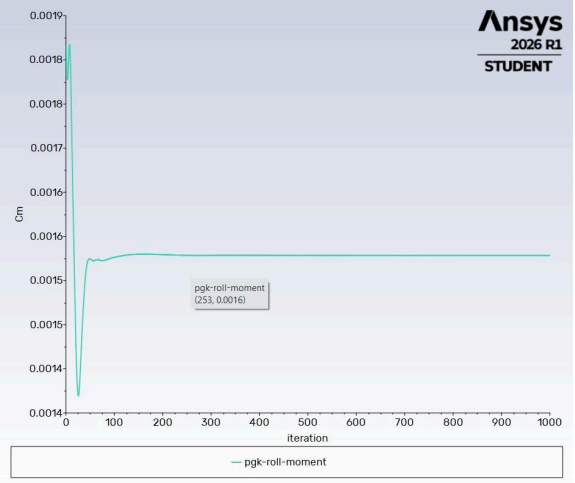


Figure 24. *PGK roll moment coefficient history, settling near 0.0016 (marker at iteration 253).*

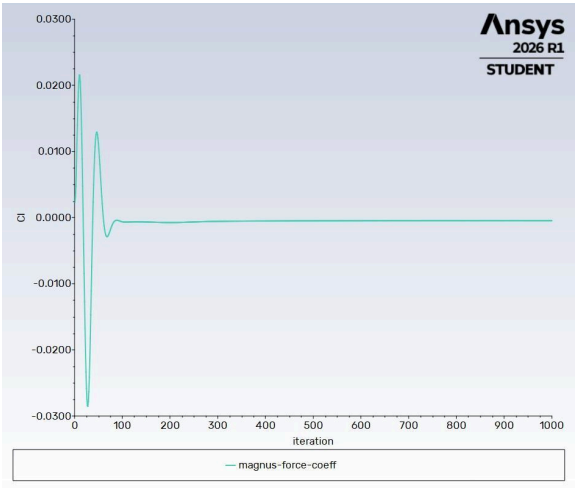


Figure 25. *Magnus force coefficient history, settling at a very small negative value.*

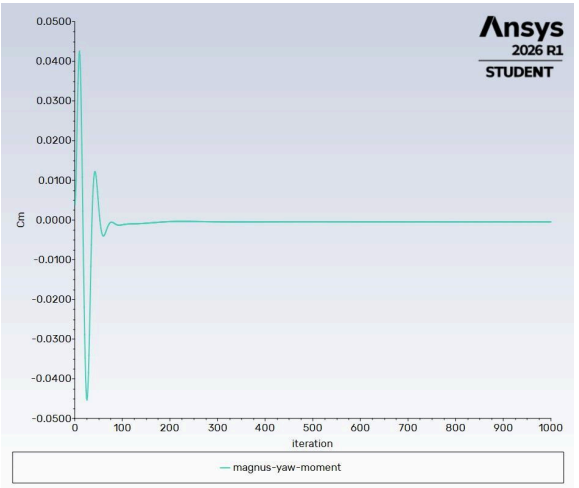


Figure 26. *Magnus yaw moment coefficient history, settling at a very small negative value.*

9.3 Converged values

Coefficient	Converged value	Reading
Drag (Cd)	about 0.39	Dominant force; includes wave drag, base drag and fin drag
Lift (Cl)	about -0.005	Very small, so the flow is close to zero incidence
Pitching moment (Cm)	about 0.016	Small; sign depends on the moment reference point and convention
PGK roll moment	about 0.0016	Small but non-zero, linked to the fin geometry and any spin
Magnus force	about -5e-4	About three orders of magnitude below Cd
Magnus yaw moment	about -5e-4	Negligible at this condition